Continuum monitoring ontology

Complete schema atlas

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Companion to the simplified and complete conceptual graphs. This atlas keeps every declared class and property readable at normal print size. It describes the asserted TBox, including the benchmark deployment extension and external vocabulary declarations, not individual users or experimental events.

Schema inventory	Count
Named classes	146
Anonymous class expressions	11
Object properties	165
Datatype properties	98
Annotation properties	14
Subclass axioms	70

Reading the diagrams

Solid arrows are object-property domain-to-range projections, not mandatory links, cardinalities or a workflow. Hollow arrows point from a subclass to its parent. OR identifies an explicit union; multiple asserted domains or ranges are combined with AND. "Not declared" means the source supplies no constraint; no class is invented to fill that gap. Datatype properties are attributes. Annotation properties and additional schema axioms are listed separately.

The simplified graph is a deliberately selected subset. The full map is a zoomable vector supplement, not a figure to shrink onto one paper column. Colors identify modules; names, borders and line styles retain meaning in grayscale. Class/individual punning is marked explicitly. SPARQL and SHACL enforcement is not inferred from a diagram.

Foundation

4 named classes | 2 properties

Classes and asserted parents

Class	subClassOf	English label
ex:MAPESymptom	-	MAPE-K Symptom
ex:NonUser	foaf:Person	Non-User
ex:ProcessingLevel	-	Processing Level
ex:User	foaf:Person	User

Object properties

Property	Asserted domain	Asserted range
ex:hasDeviceState	ex:Wearable	ex:DeviceState
ex:hasMobileDevice	ex:User	ex:MobileDevice

Topology

16 named classes | 14 properties

Classes and asserted parents

Class	subClassOf	English label
ex:CityArea	ex:UrbanEnvironment	City Area
ex:CloudNode	ex:ComputationalNode	Cloud Node
ex:Community	-	Community
ex:ComputationalNode	saref:Device	Computational Node
ex:DistanceLevel	-	Distance Level
ex:EdgeNode	ex:ComputationalNode	Edge Node
ex:FogNode	ex:ComputationalNode	Fog Node
ex:MistNode	ex:ComputationalNode	Mist Node
ex:MobileDevice	ex:MistNode	Mobile Device
ex:NodeUserRelation	ex:TemporalEntity	Node-User Relation
ex:PopulationDensity	-	Population Density
ex:RestrictedZone [punned]	ex:CityArea	Restricted Zone
ex:RuralZone [punned]	ex:CityArea	Rural Zone
ex:UrbanEnvironment	-	Urban Environment
ex:UrbanZone [punned]	ex:CityArea	Urban Zone
ex:ZoneType	-	Zone type

Object properties

Property	Asserted domain	Asserted range
ex:belongsToCommunity	ex:User	ex:Community
ex:connectsTo	saref:Device	ex:ComputationalNode
ex:coversArea	ex:ComputationalNode	ex:CityArea
ex:hasDistance	ex:NodeUserRelation	ex:DistanceLevel
ex:hasGeometry	ex:CityArea	geo:Geometry
ex:hasNeighborNode	ex:ComputationalNode	ex:ComputationalNode
ex:hasPopulationDensity	ex:CityArea	ex:PopulationDensity
ex:locatedIn	not declared	not declared
ex:locatedInZone	ex:User	ex:CityArea
ex:relatesNode	ex:NodeUserRelation	ex:ComputationalNode
ex:relatesUser	ex:NodeUserRelation	ex:User

Datatype properties

Property	Asserted domain	Asserted range
ex:continuumLevel	ex:ModelTier	xsd:integer
ex:hasElasticity	ex:ComputationalNode	xsd:boolean
ex:isPreferredNode	ex:NodeUserRelation	xsd:boolean

Observability

20 named classes | 35 properties

Classes and asserted parents

Class	subClassOf	English label
ex:AvailabilityLevel	-	Availability Level
ex:BatteryLevel	-	Battery Level
ex:CommunicationLevel	-	Communication Level
ex:ComplianceMetric	-	Compliance Metric
ex:DataContext	ex:TemporalEntity	Data Context
ex:DeviceConnectionStatus	-	Device Connection Status
ex:DeviceState	ex:State	Device State
ex:EnergyLevel	-	Energy Level
ex:MobilityLevel	-	Mobility Level
ex:NodeState	ex:State	Node State
ex:OperationalStatus	-	Operational Status
ex:PerformanceLevel	-	Performance Level
ex:ResidualCapacity	-	Residual Capacity
ex:State	ex:TemporalEntity	State
ex:TemporalEntity	-	Temporal Entity
ex:TrafficWindow	-	Traffic Window
ex:TrustAssessment	ex:TemporalEntity	Trust Assessment
ex:TrustEvidence	-	Trust Evidence
ex:UserState	ex:State	User State
ex:WorkloadLevel	-	Workload Level

Object properties

Property	Asserted domain	Asserted range
ex:contextDeviceState	ex:DataContext	ex:DeviceState
ex:contextNodeState	ex:DataContext	ex:NodeState
ex:contextProcessingLevel	ex:DataContext	ex:ProcessingLevel
ex:contextPurpose	ex:DataContext	ex:ProcessingPurpose
ex:contextZone	ex:DataContext	ex:CityArea
ex:hasAvailability	ex:NodeState	ex:AvailabilityLevel
ex:hasBatteryLevel	ex:DeviceState	ex:BatteryLevel
ex:hasCommunication	ex:NodeState	ex:CommunicationLevel
ex:hasComplianceMetric	ex:ValidationCampaign	ex:ComplianceMetric
ex:hasConnectionStatus	ex:DeviceState	ex:DeviceConnectionStatus
ex:hasDataContext	not declared	ex:DataContext
ex:hasEnergyLevel	ex:NodeState	ex:EnergyLevel
ex:hasMobility	ex:UserState	ex:MobilityLevel
ex:hasNodeState	ex:ComputationalNode	ex:NodeState
ex:hasOperationalStatus	ex:NodeState	ex:OperationalStatus
ex:hasPerformance	ex:NodeState	ex:PerformanceLevel

Property	Asserted domain	Asserted range
ex:hasResidualCapacity	ex:NodeState	ex:ResidualCapacity
ex:hasTrustAssessment	ex:NodeState	ex:TrustAssessment
ex:hasTrustEvidence	ex:TrustAssessment	ex:TrustEvidence
ex:hasUserState	ex:User	ex:UserState
ex:hasWorkload	ex:NodeState	ex:WorkloadLevel
ex:trustAssessmentForState	ex:TrustAssessment	ex:NodeState

Datatype properties

Property	Asserted domain	Asserted range
ex:evidenceContribution	ex:TrustEvidence	xsd:decimal
ex:evidenceFactor	ex:TrustEvidence	xsd:string
ex:hasTrustScore	not declared	xsd:decimal
ex:parametrizedDataReady	ex:DeviceState	xsd:boolean
ex:queuedRequests	ex:NodeState	xsd:integer
ex:resourceUsagePercent	ex:NodeState	xsd:decimal
ex:stateDuration	ex:State	xsd:duration
ex:trustRuleVersion	ex:TrustAssessment	xsd:string
ex:trustWindowEnd	ex:TrustAssessment	xsd:dateTime
ex:trustWindowStart	ex:TrustAssessment	xsd:dateTime
ex:userFeedbackScore	ex:UserState	xsd:integer
ex:validFrom	ex:TemporalEntity	xsd:dateTime
ex:validTo	ex:TemporalEntity	xsd:dateTime

Governance

44 named classes | 93 properties

Classes and asserted parents

Class	subClassOf	English label
ex:AcceptanceProfile	ex:Artifact	Acceptance Profile
ex:AnonymousIdentifier	ex:Identifier	Anonymous Identifier
ex:Artifact	-	Artifact
ex:AuthorizationDecision	ex:TemporalEntity	Authorization Decision
ex:AuthorizationOutcome	-	Authorization Outcome
ex:ConflictResolutionStrategy	-	Conflict Resolution Strategy
ex:ConsentRange	-	Consent Range
ex:ConsentRecord	ex:TemporalEntity	Consent Record
ex:DataCategory	-	Data Category
ex:DataCriticality	-	Data Criticality
ex:DataSensitivity	-	Data Sensitivity
ex:DifferentialPrivacyMechanism	ex:PrivacyMechanism	Differential Privacy Mechanism
ex:DirectIdentifier	ex:Identifier	Direct Identifier
ex:EncryptionMechanism	ex:SecurityMechanism	Encryption Mechanism
ex:FunctionalRequirement	ex:Requirement	Functional Requirement
ex:Identifier	-	Identifier
ex:MechanismSpecification	-	Mechanism Specification
ex:NonFunctionalRequirement	ex:Requirement	Non-functional Requirement
ex:OntologyArtifact	ex:Artifact	Ontology Artifact
ex:Permission	-	Permission
ex:Policy	-	Policy
ex:PolicyArtifact	ex:Artifact	Policy Artifact
ex:PolicyCategory	-	Policy Category
ex:PolicyCategoryRelation	-	Policy Category Relation
ex:PolicyRelationType	-	Policy Relation Type
ex:PolicyType	-	Policy Type
ex:PrivacyMechanism	-	Privacy Mechanism
ex:ProcessingPurpose	-	Processing Purpose
ex:ProcessingScope	-	Processing Scope
ex:PseudonymousIdentifier	ex:Identifier	Pseudonymous Identifier
ex:QueryCatalog	ex:Artifact	Query Catalog
ex:QuerySpecification	-	Query Specification
ex:QueryType	-	Query Type
ex:Requirement	-	Requirement
ex:RequirementsArtifact	ex:Artifact	Requirements Artifact
ex:Role	-	Access Role
ex:Scenario	-	Scenario
ex:ScenarioArtifact	ex:Artifact	Scenario Artifact
ex:SecurityMechanism	-	Security Mechanism
ex:SemanticContract	ex:TemporalEntity	Semantic Contract

Class	subClassOf	English label
ex:StandardSpecification	-	Standard Specification
ex:ValidationCampaign	ex:Artifact	Validation Campaign
ex:ValidationRequirement	ex:Requirement	Validation / Reproducibility Requirement
ex:ValidationResult	-	Validation Result

Object properties

Property	Asserted domain	Asserted range
ex:appliedMechanism	ex:EvaluationState	ex:MechanismSpecification
ex:appliedPolicy	ex:EvaluationState	ex:Policy
ex:appliedSecurityMechanism	not declared	ex:SecurityMechanism
ex:appliesInZone	ex:Policy	ex:CityArea
ex:appliesTo	ex:Permission	not declared
ex:appliesToZoneType	ex:Policy	ex:ZoneType
ex:basedOnConsentRecord	ex:AuthorizationDecision	ex:ConsentRecord
ex:basedOnContract	ex:AuthorizationDecision	ex:SemanticContract
ex:basedOnZone	ex:AuthorizationDecision	ex:CityArea
ex:belongsToPolicyCategory	ex:Policy	ex:PolicyCategory
ex:belongsToQueryCatalog	ex:QuerySpecification	ex:QueryCatalog
ex:consentSubject	ex:ConsentRecord	ex:User
ex:contractSubject	ex:SemanticContract	ex:User
ex:definedInArtifact	not declared	ex:Artifact
ex:governedBy	not declared	ex:Policy
ex:hasActiveConsentRange	ex:User	ex:ConsentRange
ex:hasAuthorizationDecision	not declared	ex:AuthorizationDecision
ex:hasAuthorizationOutcome	ex:AuthorizationDecision	ex:AuthorizationOutcome
ex:hasAuthorizedDataCategory	ex:ConsentRecord	ex:DataCategory
ex:hasConsentRange	(ex:User OR ex:SemanticContract OR ex:Policy OR ex:FederatedLearningSessio n OR ex:EvaluationState)	ex:ConsentRange
ex:hasConsentRecord	ex:User	ex:ConsentRecord
ex:hasDataCategory	not declared	ex:DataCategory
ex:hasDataCriticality	not declared	ex:DataCriticality
ex:hasDataSensitivity	not declared	ex:DataSensitivity
ex:hasEffectiveConsentRange	ex:AuthorizationDecision	ex:ConsentRange
ex:hasIdentifier	ex:User	ex:Identifier
ex:hasPermission	ex:Role	ex:Permission
ex:hasPolicyRelationType	ex:PolicyCategoryRelation	ex:PolicyRelationType
ex:hasPolicyType	ex:Policy	ex:PolicyType
ex:hasProcessingPurpose	(ex:SemanticContract OR ex:Policy OR ex:FederatedLearningSession OR ex:EvaluationState OR ex:Permission)	ex:ProcessingPurpose
ex:hasProcessingScope	ex:ModelTier	ex:ProcessingScope
ex:hasQueryType	ex:QuerySpecification	ex:QueryType
ex:hasRole	foaf:Person	ex:Role
ex:hasSemanticContract	ex:User	ex:SemanticContract
ex:hasValidationResult	ex:ValidationCampaign	ex:ValidationResult

Property	Asserted domain	Asserted range
ex:policyArtifactUsed	ex:EvaluationState	ex:PolicyArtifact
ex:recommendedMechanism	ex:Policy	ex:MechanismSpecification
ex:relatedRequirement	ex:Policy	ex:Requirement
ex:relationSourceCategory	ex:PolicyCategoryRelation	ex:PolicyCategory
ex:relationTargetCategory	ex:PolicyCategoryRelation	ex:PolicyCategory
ex:requirementsArtifactUsed	ex:EvaluationState	ex:RequirementsArtifact
ex:requiresConsentRange	(ex:Permission OR ex:Policy OR ex:Service OR ex:ServiceState OR ex:AIModel OR ex:FederatedLearningSession OR ex:EvaluationState)	ex:ConsentRange
ex:scenarioMechanism	ex:Scenario	ex:MechanismSpecification
ex:scenarioPolicy	ex:Scenario	ex:Policy
ex:supportsPolicy	ex:MechanismSpecification	ex:Policy
ex:tracedToMechanism	ex:Requirement	ex:MechanismSpecification
ex:tracedToPolicy	ex:Requirement	ex:Policy
ex:usesAcceptanceProfile	ex:ValidationCampaign	ex:AcceptanceProfile
ex:usesIdentifier	ex:TransferEvent	ex:Identifier
ex:usesResolutionStrategy	ex:PolicyCategoryRelation	ex:ConflictResolutionStrategy
ex:usesStandard	ex:Artifact	ex:StandardSpecification
ex:validationUsesOntology	ex:ValidationCampaign	ex:OntologyArtifact
ex:validationUsesPolicies	ex:ValidationCampaign	ex:PolicyArtifact
ex:validationUsesQueries	ex:ValidationCampaign	ex:QueryCatalog
ex:validationUsesScenarios	ex:ValidationCampaign	ex:ScenarioArtifact

Datatype properties

Property	Asserted domain	Asserted range
ex:AHP_consistency_threshold	ex:AcceptanceProfile	xsd:decimal
ex:D_delegation_max	ex:AcceptanceProfile	xsd:integer
ex:E_device_max	ex:AcceptanceProfile	xsd:decimal
ex:N_agents	ex:AcceptanceProfile	xsd:integer
ex:T_decision_max	ex:AcceptanceProfile	xsd:decimal
ex:T_inference_local	ex:AcceptanceProfile	xsd:decimal
ex:T_migration_max	ex:AcceptanceProfile	xsd:decimal
ex:T_node_join	ex:AcceptanceProfile	xsd:decimal
ex:T_reselection_max	ex:AcceptanceProfile	xsd:decimal
ex:T_sparql_monitor	ex:AcceptanceProfile	xsd:decimal
ex:artifactIdentifier	ex:Artifact	xsd:string
ex:artifactStatus	ex:Artifact	xsd:string
ex:artifactVersion	ex:Artifact	xsd:string
ex:authorizationReason	ex:AuthorizationDecision	xsd:string
ex:categoryCode	ex:PolicyCategory	xsd:string
ex:conditionExpression	ex:RecoveryCondition	xsd:string
ex:configurationStatus	not declared	xsd:string
ex:containsPersistentIdentifier	ex:FederatedLearningSession	xsd:boolean
ex:containsPersonalData	ex:FederatedLearningSession	xsd:boolean

Property	Asserted domain	Asserted range
ex:hasPolicyAction	ex:Policy	xsd:string
ex:hasPolicyStatement	ex:Policy	rdfs:Literal
ex:identifierScope	ex:Identifier	xsd:string
ex:identifierValue	ex:Identifier	xsd:string
ex:mechanismDescription	ex:MechanismSpecification	rdfs:Literal
ex:mechanismIdentifier	ex:MechanismSpecification	xsd:string
ex:policyIdentifier	ex:Policy	xsd:string
ex:policyVersion	ex:Policy	xsd:string
ex:protectsAtRest	ex:SecurityMechanism	xsd:boolean
ex:protectsInTransit	ex:SecurityMechanism	xsd:boolean
ex:queryIdentifier	ex:QuerySpecification	xsd:string
ex:queryInterpretation	ex:QuerySpecification	xsd:string
ex:queryPurpose	ex:QuerySpecification	xsd:string
ex:relationCode	ex:PolicyRelationType	xsd:string
ex:requirementIdentifier	ex:Requirement	xsd:string
ex:requirementStatement	ex:Requirement	xsd:string
ex:scenarioIdentifier	ex:Scenario	xsd:string
ex:securityBaselineVersion	ex:SecurityMechanism	xsd:string
ex:strategyCode	ex:ConflictResolutionStrategy	xsd:string

Orchestration

28 named classes | 79 properties

Classes and asserted parents

Class	subClassOf	English label
ex:AIModel	-	AI Model
ex:AdaptabilityLevel	-	Adaptability Level
ex:AdaptationAction	ex:TemporalEntity	Adaptation Action
ex:BufferRecord	ex:TemporalEntity	Buffer Record
ex:DecisionAlternative	-	Decision Alternative
ex:DecisionMethod	-	Decision Method
ex:DegradationEvent	ex:AdaptationAction	Degradation Event
ex:DelegationEvent	ex:AdaptationAction ex:TemporalEntity	Delegation Event
ex:EvaluationState	ex:State	Evaluation State
ex:MigrationCostLevel	-	Migration Cost Level
ex:MigrationEvent	ex:AdaptationAction	Migration Event
ex:MigrationTimeLevel	-	Migration Time Level
ex:ModelDegradationCause	-	Model Degradation Cause
ex:ModelSelectionAction	ex:AdaptationAction	Model Selection Action
ex:ModelTier	-	Model Tier
ex:OffloadingEvent	ex:AdaptationAction	Offloading Event
ex:PairwiseComparison	-	Pairwise Comparison
ex:ParametrizedData	-	Parametrized Data
ex:ProfitabilityLevel	-	Profitability Level
ex:RecoveryCondition	-	Recovery Condition
ex:ReplicationEvent	ex:TemporalEntity	Replication Event
ex:RetentionEvent	ex:AdaptationAction	Retention Event
ex:RollbackEvent	ex:AdaptationAction	Rollback Event
ex:ScalingEvent	ex:AdaptationAction	Scaling Event
ex:Service	-	Service
ex:ServiceState	ex:State	Service State
ex:SynchronizationEvent	ex:AdaptationAction	Synchronization Event
ex:TransferEvent	ex:TemporalEntity	Transfer Event

Object properties

Property	Asserted domain	Asserted range
ex:actionOriginNode	ex:AdaptationAction	ex:ComputationalNode
ex:actionTargetNode	ex:AdaptationAction	ex:ComputationalNode
ex:affectsModel	ex:AdaptationAction	ex:AIModel
ex:affectsService	ex:AdaptationAction	ex:Service
ex:alternativeModelTier	ex:DecisionAlternative	ex:ModelTier
ex:appliesPolicy	ex:EvaluationState	ex:Policy
ex:auditsContract	ex:EvaluationState	ex:SemanticContract
ex:authorizedByEvaluation	ex:AdaptationAction	ex:EvaluationState

Property	Asserted domain	Asserted range
ex:carriesData	ex:TransferEvent	not declared
ex:criterionA	ex:PairwiseComparison	not declared
ex:criterionB	ex:PairwiseComparison	not declared
ex:delegatedBy	ex:DelegationEvent	ex:ComputationalNode
ex:delegatesTo	(ex:ComputationalNode OR ex:DelegationEvent)	ex:ComputationalNode
ex:derivedFrom	ex:State	sosa:Observation
ex:evaluatesNode	ex:EvaluationState	ex:ComputationalNode
ex:evaluatesService	ex:EvaluationState	ex:Service
ex:evaluationPurpose	ex:EvaluationState	ex:ProcessingPurpose
ex:evaluationUser	ex:EvaluationState	ex:User
ex:evaluationZone	ex:EvaluationState	ex:CityArea
ex:expiresOnRecoveryOf	ex:DelegationEvent	ex:ComputationalNode
ex:hasAdaptability	ex:ServiceState	ex:AdaptabilityLevel
ex:hasDecisionAlternative	ex:EvaluationState	ex:DecisionAlternative
ex:hasDecisionMethod	ex:EvaluationState	ex:DecisionMethod
ex:hasDegradationCause	not declared	ex:ModelDegradationCause
ex:hasDelegation	ex:ComputationalNode	ex:DelegationEvent
ex:hasDetectedSymptom	ex:EvaluationState	ex:MAPESymptom
ex:hasMigrationCost	ex:EvaluationState	ex:MigrationCostLevel
ex:hasMigrationTime	ex:EvaluationState	ex:MigrationTimeLevel
ex:hasModelTier	ex:AIModel	ex:ModelTier
ex:hasPairwiseComparison	ex:EvaluationState	ex:PairwiseComparison
ex:hasProcessingLevel	ex:ServiceState	ex:ProcessingLevel
ex:hasProfitability	ex:EvaluationState	ex:ProfitabilityLevel
ex:hasRecoveryCondition	ex:DelegationEvent	ex:RecoveryCondition
ex:hasReplicationEvent	not declared	ex:ReplicationEvent
ex:hasServiceState	ex:Service	ex:ServiceState
ex:hostedOnNode	ex:AIModel	ex:ComputationalNode
ex:hostsModel	ex:ComputationalNode	ex:AIModel
ex:hostsService	ex:ComputationalNode	ex:Service
ex:parentDelegation	ex:DelegationEvent	ex:DelegationEvent
ex:partOfScenario	not declared	ex:Scenario
ex:replicaOf	ex:DeploymentFragment	ex:DeploymentFragment
ex:replicationSource	ex:ReplicationEvent	not declared
ex:replicationTarget	ex:ReplicationEvent	not declared
ex:resultedInAction	ex:EvaluationState	ex:AdaptationAction
ex:rollbackTarget	ex:RollbackEvent	ex:AIModel
ex:selectedAlternative	ex:EvaluationState	ex:DecisionAlternative
ex:selectedModelTier	ex:EvaluationState	ex:ModelTier
ex:supersedesModel	ex:AIModel	ex:AIModel
ex:transferDestination	ex:TransferEvent	not declared
ex:transferSource	ex:TransferEvent	not declared
ex:triggeredByState	(ex:DelegationEvent OR ex:EvaluationState)	ex:State

Property	Asserted domain	Asserted range
ex:usesModel	ex:Service	ex:AIModel

Datatype properties

Property	Asserted domain	Asserted range
ex:delegationDepth	ex:DelegationEvent	xsd:integer
ex:eligibilityReason	ex:DecisionAlternative	xsd:string
ex:estimatedPredictionError	ex:EvaluationState	xsd:decimal
ex:evaluationOrder	ex:PolicyCategory	xsd:integer
ex:hasAHPScore	ex:DecisionAlternative	xsd:decimal
ex:hasConsistencyRatio	ex:EvaluationState	xsd:decimal
ex:hasConsistencyThreshold	ex:EvaluationState	xsd:decimal
ex:hasEvaluationTicketID	ex:EvaluationState	xsd:string
ex:hasLatencyWeight	ex:EvaluationState	xsd:decimal
ex:hasModelQualityWeight	ex:EvaluationState	xsd:decimal
ex:hasPrivacyWeight	ex:EvaluationState	xsd:decimal
ex:hasSelectionJustification	ex:EvaluationState	xsd:string
ex:hasTrustWeight	ex:EvaluationState	xsd:decimal
ex:idempotencyKey	not declared	xsd:string
ex:isEligible	ex:DecisionAlternative	xsd:boolean
ex:lastUpdated	ex:AIModel	xsd:dateTime
ex:legacyDecisionScore	ex:EvaluationState	xsd:decimal
ex:migrationNote	not declared	xsd:string
ex:migrationStatus	not declared	xsd:string
ex:modelLineageStatus	ex:AIModel	xsd:string
ex:modelVersion	ex:AIModel	xsd:string
ex:observedModelQuality	ex:EvaluationState	xsd:decimal
ex:pairwiseValue	ex:PairwiseComparison	xsd:decimal
ex:plannedExpiry	ex:TemporalEntity	xsd:dateTime
ex:predictionConfidence	ex:EvaluationState	xsd:decimal
ex:replicationVersion	not declared	xsd:string
ex:requiresRecalculation	ex:EvaluationState	xsd:boolean

Federated learning

4 named classes | 22 properties

Classes and asserted parents

Class	subClassOf	English label
ex:FederatedLearningSession	-	Federated Learning Session
ex:ModelGradientUpdate	ex:ParametrizedData	Model Gradient Update
ex:PayloadType	-	Payload Type
ex:PrivacyBudgetAccount	-	Privacy Budget Account

Object properties

Property	Asserted domain	Asserted range
ex:aggregatesData	ex:ComputationalNode	ex:ParametrizedData
ex:budgetForContract	ex:PrivacyBudgetAccount	ex:SemanticContract
ex:budgetForPurpose	ex:PrivacyBudgetAccount	ex:ProcessingPurpose
ex:hasPayloadType	ex:FederatedLearningSession	ex:PayloadType
ex:hasPrivacyBudgetAccount	ex:FederatedLearningSession	ex:PrivacyBudgetAccount
ex:hasPrivacyMechanism	(ex:FederatedLearningSession OR ex:ParametrizedData)	ex:PrivacyMechanism
ex:involvedNode	ex:FederatedLearningSession	ex:ComputationalNode
ex:originatesFromDevice	ex:ParametrizedData	ex:MobileDevice
ex:sentToNode	ex:ParametrizedData	ex:ComputationalNode
ex:transfersData	ex:FederatedLearningSession	ex:ParametrizedData
ex:updatesModel	ex:FederatedLearningSession	ex:AIModel

Datatype properties

Property	Asserted domain	Asserted range
ex:containsIndividualizedGradients	ex:FederatedLearningSession	xsd:boolean
ex:hasAnonymizationApplied	ex:ParametrizedData	xsd:boolean
ex:hasNoiseApplied	ex:ParametrizedData	xsd:boolean
ex:hasPrivacyBudget	ex:FederatedLearningSession	xsd:decimal
ex:isRedundant	ex:ParametrizedData	xsd:boolean
ex:noiseLevel	ex:FederatedLearningSession	xsd:decimal
ex:privacyBudgetConsumed	ex:PrivacyBudgetAccount	xsd:decimal
ex:privacyBudgetMaximum	ex:PrivacyBudgetAccount	xsd:decimal
ex:privacyBudgetRemaining	ex:PrivacyBudgetAccount	xsd:decimal
ex:sendTimestamp	ex:ParametrizedData	xsd:dateTime
ex:sessionTime	ex:FederatedLearningSession	xsd:dateTime

Deployment

4 named classes | 6 properties

Classes and asserted parents

Class	subClassOf	English label
ex:AuthorityScope	-	Authority scope
ex:DeploymentFragment	-	Deployment fragment
ex:PrivacyClass	-	Privacy class
ex:SemanticNodeProfile	-	Semantic node profile

Object properties

Property	Asserted domain	Asserted range
ex:authoritativeFor	ex:DeploymentFragment	ex:AuthorityScope
ex:hasPrivacyClass	ex:DeploymentFragment	ex:PrivacyClass
ex:hostedAt	ex:DeploymentFragment	ex:ComputationalNode

Datatype properties

Property	Asserted domain	Asserted range
ex:graphIRI	ex:DeploymentFragment	xsd:anyURI
ex:ontologyDigest	ex:SemanticNodeProfile	xsd:string
ex:usesEntailmentProfile	ex:SemanticNodeProfile	xsd:string

Wellbeing (optional)

18 named classes | 4 properties

Classes and asserted parents

Class	subClassOf	English label
ex:AccelerometerSensor	ex:PhysiologicalSensor	Accelerometer Sensor
ex:EDASensor	ex:PhysiologicalSensor	EDA Sensor
ex:HeartRateSensor	ex:PhysiologicalSensor	Heart Rate Sensor
ex:PersonStatus	-	Person Status
ex:PhysiologicalObservation	sosa:Observation	Physiological Observation
ex:PhysiologicalParametrizedData	ex:ParametrizedData	Physiological Parametrized Data
ex:PhysiologicalSensor	sosa:Sensor	Physiological Sensor
ex:SleepObservation	sosa:Observation	Sleep Observation
ex:SleepParametrizedData	ex:ParametrizedData	Sleep Parametrized Data
ex:SleepSensor	ex:PhysiologicalSensor	Sleep Sensor
ex:SmartBand	ex:Wearable	Smart Band
ex:SmartRing	ex:Wearable	Smart Ring
ex:SmartWatch	ex:Wearable	Smart Watch
ex:SpO2Sensor	ex:PhysiologicalSensor	SpO2 Sensor
ex:StressLevel	-	Stress Level
ex:StressObservation	sosa:Observation	Stress Observation
ex:TemperatureSensor	ex:PhysiologicalSensor	Skin Temperature Sensor
ex:Wearable	saref:Device	Wearable

Object properties

Property	Asserted domain	Asserted range
ex:generatedBy	ex:ParametrizedData	ex:Wearable
ex:hasPersonStatus	ex:UserState	ex:PersonStatus
ex:hasPredictedStressLevel	ex:UserState	ex:StressLevel
ex:hasWearable	ex:User	ex:Wearable

External standards / SHACL

8 named classes | 22 properties

Classes and asserted parents

Class	subClassOf	English label
foaf:Person	-	Person
geo:Feature	-	Feature
geo:Geometry	-	Geometry
saref:Device	-	Device
sh:NodeShape	-	Node Shape
sosa:Observation	-	Observation
sosa:Platform	-	Platform
sosa:Sensor	-	Sensor

Object properties

Property	Asserted domain	Asserted range
geo:hasGeometry	not declared	not declared
sosa:hasFeatureOfInterest	not declared	not declared
sosa:isHostedBy	not declared	not declared
sosa:madeBySensor	not declared	not declared
sosa:observedProperty	not declared	not declared

Datatype properties

Property	Asserted domain	Asserted range
foaf:name	not declared	not declared
sosa:hasSimpleResult	not declared	not declared
sosa:resultTime	not declared	not declared

Annotation properties

Property	Asserted domain	Asserted range
dcterms:identifier	not declared	not declared
dcterms:language	not declared	not declared
dcterms:modified	not declared	not declared
dcterms:replaces	not declared	not declared
sh:class	not declared	not declared
sh:datatype	not declared	not declared
sh:maxCount	not declared	not declared
sh:message	not declared	not declared
sh:minCount	not declared	not declared
sh:or	not declared	not declared
sh:path	not declared	not declared
sh:property	not declared	not declared
sh:severity	not declared	not declared
sh:targetClass	not declared	not declared

Additional axioms and provenance

Subject	Axiom	Object
ex:hasGeometry	subproperty	geo:hasGeometry

Full IRIs, expression members, declaration source files and comments are retained in data/schema.json and the CSV inventories. GraphML retains term identity and schema edges for graph editors, not the full OWL model. No additional ontology is imported from the network. The hashes below identify the exact 14 runtime inputs, including the deployment extension.

Source	SHA-256
ontology/core/schema.ttl	9f113738ed7a39ec42b134766078378e9282daf955469778b689a8d3720923cd
ontology/modules/foundation.ttl	dbb535b2e1f3b6a13a4c7e985a6193c5f7beb356e15f5fe3c6cf16272f3fcb48
ontology/modules/topology.ttl	a83db2c195864846a9bc029ddc0b75d2e37ad860354e6d8ec25fdaf60953d9cd
ontology/modules/observability.ttl	307d301bb86a7186000b5b6dd3073a21ddd9834a17cec31218d8b9fc7c61584e
ontology/modules/governance.ttl	e89c189627bd6deb433bc8de6d351fc4514f8ee1510a90cab94cf75e826f4a06
ontology/modules/orchestration.ttl	9381540d925396f7efb47b207e1fc88cdd657669f505483116ba31c3087d6b29
ontology/modules/federation.ttl	0bd7b9288b77b054208d7e1d287f365a8aedca9f77f0a46a6cc905cb49ca5630
ontology/modules/deployment.ttl	3d35c0e51a1d2dd4da5eb83716b1f7bbbf0768a4bf256265d4dadabcaf53a1c0
ontology/core/vocabulary.ttl	2b6412b5ebaf7f5d3efa4de9ad42e4bcfd26d4c6714fd4a3f7666cdcf9ff9d40
ontology/domains/wellbeing/schema.ttl	26a7bcb85c5b1653f040a1be82d43576ca41d84c6b3008766ec3d68d0453540c
ontology/domains/wellbeing/vocabulary.ttl	ba855f89ab49b42ed2dd13a21782c616b10b7f5db3481c0376f31e58c9e92590
ontology/examples/reference-system.ttl	a29d8e2bce486836d42f566662adcc3ba098dc5bbb877e7ae4fad32399a83683
ontology/shapes/core-compliance.ttl	e3fa7e3a58199022e62756e7b145659acc0bf2f8be0f03b63625726298e3ad41
ontology/shapes/federated-learning.ttl	d5d705a937bb7fa9a58391c48dd6253fe98078af7af8df96f27169a9b0cb6652